Operating Instructions

Diesel engine-generator set
mtu 6R1600 DS300 / DG06R1600A1E
mtu 6R1600 DS330 / DG06R1600A2E

Controller MM7, MC7

SS180022/05E

A Rolls-Royce solution

System designation	Engine model	Application group
mtu 6R1600 DS300 / DG06R1600A1E	6R1600G10F	3B, continuous service, variable load, ICXN (PRP)
mtu 6R1600 DS300 / DG06R1600A1E	6R1600G70F	3D, emergency standby power, IFN (ESP)
mtu 6R1600 DS330 / DG06R1600A2E	6R1600G20F	3B, continuous service, variable load, ICXN (PRP)
mtu 6R1600 DS330 / DG06R1600A2E	6R1600G80F	3D, emergency standby power, IFN (ESP)

Table 1: Applicability

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1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

The operating instructions are divided into the following chapters:
- Safety
- Transport
- Product overview
- Technical data
- Operation
- Maintenance Schedule
- Troubleshooting
- Task description
- Monitoring and control
- Fluids and Lubricants Specifications

Store these operating instructions where they can be accessed by all persons entrusted with operating or servicing the product.

Read these Operating Instructions carefully before starting any work. For safe operation of the product, always follow all safety instructions and work instructions.

Local accident prevention regulations and general safety regulations for the area of application of the product also apply.

These Operating Instructions form part of the product and refer to the latest design version of the product at the time of publication. They are subject to modification to reflect technical enhancements. Actual product may vary from figures shown.

TIM-ID: 0000092579 - 002

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:
- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:http://www.mtu-solutions.com

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2.2 Intended use of all products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:
- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

The product must not be operated in explosive atmospheres unless the engine fulfills the conditions for such use and approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:
- http://www.mtu-solutions.com

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

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2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:
- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

TIM-ID: 0000040531 - 013

2.4 Safety regulations for commissioning and operation

Safety regulations for commissioning

Install the product correctly and carry out acceptance in accordance with the manufacturer's specifications before commissioning the product. All necessary approvals must be granted by the relevant authorities and all requirements for commissioning must be fulfilled.

Whenever the product is subsequently put into operation ensure that:
- All personnel is clear of the danger zone surrounding moving parts of the machine. Electrically-actuated linkages may be set in motion when the Engine Control Unit (governor) is switched on.
- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- All protective devices are in place.
- All lines bearing media are ready for operation.
- The walkways in the operating room are unobstructed and safe.
- No persons wearing pacemakers or any other technical body aids are present.
- Adequate ventilation of the operating room is ensured for any operating status.
- In the first few hours of operation, the product emits gases as a result of smoldering e.g. lacquers or oil. These gases may be hazardous to health. Always wear respiratory protection in the operating room during this period.
- The exhaust system is leak-tight and the gases are vented to atmosphere.
- The product is free of any damage, this applies in particular to lines and cabling.
- All components are properly grounded. Ground separately by means of a grounding stake as necessary.
- The power supply (mains plug) is always freely accessible to facilitate disconnecting the device or system from the power supply at any time.
- Battery terminals, generator terminals or cables are protected against accidental contact.
- All connections have been correctly allocated e.g. +/- polarity, fuel line/reducing agent line, supply/return.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems work properly.

Smoking is prohibited in the area of the product.

Safety regulations during operation

The operator must be familiar with the control and display elements.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

Practice emergency procedures, especially emergency stopping, at regular intervals.

Take the following steps if any system malfunctions are detected or signaled by the system:
- Inform supervisor(s) in charge.
- Analyze the message.
- Respond by taking any necessary emergency action, e.g. emergency stop.

After a safety shutdown, the engine may only be started after the cause of the shutdown has been eliminated.

Contact Service if the root cause of the malfunction cannot be clearly identified.

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Operation

Do not remain in the operating room when the product is running unless absolutely necessary. Keep your stay as short as possible.